Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part –February 16th 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the **Parallel Data Management Patterns: Pipeline**.
Are flow dependencies preventing correct parallelization of pipeline parallel pattern when present? (2)

B. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=1; i<100; i++)
        for (j=0; j<100; j++)
            b[i][j] += f(a[i][j]);
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the parallel scatter pattern: assumptions, complexity, and example. (2)

B. Please describe how the Pack parallel pattern works. How is implemented? (2)

Exercise n. 3

Answer the following questions about shared memory parallel programming and briefly explain (without an explanation, the answer will be considered invalid)

A. Why is synchronization necessary when programming with the shared memory model? (1)

B. Which Pthreads API call is needed to ensure that spawned threads can finish their execution after main() returns? (1)

C. What is the difference between static and dynamic schedule of OpenMP for loops? (1)

D. Describe what is the effect of the following Halide schedule: (1)

```
gradient.split(x, x_outer, x_inner, 4);
gradient.vectorize(x_inner);
```

Exercise n. 4

Answer the following questions about MPI and briefly explain (without an explanation, the answer will be considered invalid)

A. When is it possible that at least two MPI processes are stuck waiting for each other, without the possibility of continuing their execution? (1) **This question can be skipped if you passed the challenge.**

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B. When in doubt about the behavior of an MPI function, why is it important to rely on the MPI standard instead of just trying how it works with your MPI implementation? (1)

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C. What is the relationship between i) MPI functions with blocking semantics, ii) MPI functions without blocking semantics, and iii) MPI + multithreading? (2)

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